Anchored multi-omics integration and knowledge-anchored residual discovery: a never-below-the-best-view integrator that finds what the textbook misses

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Running title: Knowledge-anchored residual discovery

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Abstract

Background. Multi-omics integration is widely assumed to improve prediction, yet on real cancer data a strong single modality is hard to beat and naive fusion often underperforms it. **Methods.** We define an *anchored* integrator that anchors on the empirically strongest modality — or on a fixed, zero-parameter textbook prior — and adds the remaining data only as a non-negative gated residual, so the model is provably never below its anchor and improves on it only where another view carries orthogonal signal. Mining that residual yields a discovery procedure that names anchor-orthogonal features, with a matched-random-panel noise control and held-out / permutation verification. We evaluate on TCGA-BRCA (n up to 1,152) and validate externally on METABRIC (n = 1,175–1,980). **Results.** Across nine subtype labels the gate adds nothing where one modality dominates and engages only on a constructed positive control (+0.047 AUROC). Anchoring on a fixed 20-gene proliferation signature (zero trained parameters) reaches AUROC 0.919 on Luminal A vs B — nearly the trained 1,500-gene model (0.942) — and gating genome-wide data reaches 0.947, the only clean super-additive gain observed. The residual of that prior recovers a basal/keratinization axis (Reactome $p \approx 8\times10^{-11}$), verified by held-out replication (10/10 splits), stability (10/10), and a permuted-label null; it **reproduces in METABRIC** (panel Δ +0.036; independent re-discovery overlaps 20/30, hypergeometric $p \approx 7\times10^{-27}$). A second anchor (the ERBB2 amplicon for HER2) yields a distinct neuroendocrine/immune axis, and an estrogen-receptor signature yields nothing (a specificity control). The HER2 axis does **not** replicate in METABRIC because the amplicon anchor is near-complete there (0.997). The method is also not breast-cancer- or expression-specific: on an NSCLC anti-PD-1 immunotherapy cohort (n = 227, mutation/clinical features), anchored on the textbook biomarker tumour mutational burden, the residual independently recovers the field's other established biomarkers — PD-L1 (orthogonal to TMB), EGFR and STK11 mutation (resistance) (panel Δ +0.061 vs +0.006, p = 0.038). Finally, the discovered axis replicates across *cancers*: in TCGA lung (LUAD vs LUSC, n =

1,129), the same proliferation-anchored residual recovers the squamous/keratinization program, overlapping the breast basal panel 10/30 ($p \approx 3\times10^{-16}$) — the same genes in a different cancer. **Conclusion.** Anchor on established biology and let a gate decide, honestly, whether genome-wide data beats it; where it does, the residual names the new axis — one that reproduces across cohorts (METABRIC) and across cancers (lung), and generalises to a different disease and feature type (NSCLC immunotherapy). External reproducibility tracks biological coherence.

1. Introduction

The prevailing premise that adding omics layers improves prediction is contradicted by careful benchmarks: a large-scale TCGA benchmark across 14 cancers found that adding data types most often *hurt* survival prediction, with mRNA (±miRNA) sufficient for most cancers [1]; multi-modal survival models often fail to beat a small two-modality subset, so the useful combination is task-specific and narrow [2]; and naive concatenation underperforms — a principal-component baseline is hard to beat, so adaptive weighting is needed to realise any gain from fusion [3,4]. The honest objective is therefore not "always fuse" but to *never do worse than the best single modality, and improve on it only where another modality carries signal the leader cannot.* We formalise such an integrator, generalise its anchor from a data modality to established knowledge, and repurpose the construction as an interpretable discovery engine — then ask, in an independent cohort, whether its discoveries reproduce.

2. Results

An anchored gate is never below the best single view. On Luminal A vs B, RNA alone reaches AUROC 0.947 and methylation 0.745; a naive RNA+methylation stack drops to 0.941 (below RNA), whereas the gated combiner holds at 0.947 (weight $\beta = 0$ in 9/10 repeats). A constructed positive control — a held-out methylation set the RNA cannot see — engages the gate ($\beta = 4$–8) for a significant +0.047. Run blind on nine expert-defined subtype labels, anchor selection routes to methylation for all five methylation-defined clusters and to RNA for all four PAM50 calls, never below the leader: the method follows the biology rather than a built-in preference.

A zero-parameter textbook prior is a strong anchor, and yields the only clean fusion gain (Figure 1A). Luminal A vs B is, by textbook, a proliferation distinction. A 20-gene proliferation index with zero trained parameters reaches AUROC 0.919 — close to the fully trained 1,500-gene model (0.942) — and gating genome-wide data onto this fixed prior reaches 0.947 (Δ +0.029), exceeding the pure data model. With the Horvath epigenetic clock [9] as the anchor for normal-tissue age, the clock alone (0.947) already beats RNA (0.911) and data adds ≈ 0: the textbook suffices and the gate says so.

The residual names a verified new axis. Mining the proliferation-anchored residual surfaces the basal / squamous-lineage axis (KRT5/14/17/6B, TP63, DSG3/DSC3, SOX10, COL17A1, KLK5/7/8; partial $r \approx 0.46$). It is verified three ways — a train-selected panel beats random panels on a held-out test in 10/10 splits (no selection leakage), the basal core recurs in 10/10 splits, and under permuted labels the advantage collapses — and is enriched for cornified-envelope formation / keratinization / epidermis development (Reactome $p \approx 8\times10^{-11}$).

Generalisation and specificity (Figure 1A). Anchoring HER2 status on the ERBB2 amplicon (incomplete in TCGA, AUROC 0.752) discovers a distinct, verified neuroendocrine/secretory + immune axis (Δ +0.054; panel vs random $p = 0.024$; held-out 8/8). Anchoring ER status on a textbook ER/luminal signature (complete, 0.938) discovers nothing (Δ −0.001): a real-data specificity control showing the method does not manufacture axes.

External validation (Figure 1B). In the independent METABRIC cohort the basal axis reproduces: the fixed TCGA panel adds Δ +0.036 over the proliferation prior (combined 0.960; beats random panels, $p = 0.048$), and an unbiased re-discovery independently recovers the same basal genes, overlapping the TCGA panel 20/30 (hypergeometric $p \approx 7\times10^{-27}$). The HER2 axis does **not** reproduce — there the amplicon anchor is near-complete (0.997 vs 0.752 in TCGA), leaving no residual (panel $\Delta \approx 0$; overlap 2/30). External reproducibility tracks biological coherence: the pathway-enriched basal axis reproduces, while the diffuse HER2 axis was a cohort-specific residual.

Cross-domain generalisation (a different cancer, question, and feature type). The method is not specific to breast cancer or to expression. On NSCLC patients receiving anti-PD-1 checkpoint blockade (Hellmann/MSK 2018, n = 227; mutation and clinical features; endpoint = durable clinical benefit), anchoring on the textbook immuno-oncology biomarker tumour mutational burden (TMB; anchor AUROC 0.60) and mining the residual independently recovers the field's *other* established biomarkers: PD-L1 score (positive partial correlation, and genuinely orthogonal to TMB, corr 0.00), EGFR mutation and STK11/LKB1 mutation (both negative — known checkpoint-blockade resistance). The discovered panel adds Δ +0.061 versus +0.006 for matched random panels ($p = 0.038$). Anchored on the textbook biomarker, the procedure rediscovers the known complementary biomarkers of a different disease.

Cross-cancer replication and tissue-independence of the basal axis. The strongest test that a discovered axis is real biology, not a cohort artefact, is to seek it in a *different cancer*. In TCGA lung (LUAD adeno vs LUSC squamous, n = 1,129), the same proliferation-anchored residual recovers the squamous/keratinization program and overlaps the breast basal panel 10/30 (hypergeometric $p \approx 3\times10^{-16}$) — the same genes, a different cancer. (On this near-trivial histology split the panel-vs-random margin saturates, so gene-level overlap is the informative metric.) Adding head & neck (HNSC), the breast panel separates squamous (HNSC + LUSC) from adeno (LUAD) at AUROC 0.96 with HNSC and LUSC — two different tissues — both scoring high and LUAD low, and within HNSC the score tracks differentiation grade (G1 > G3, $p \approx 2\times10^{-4}$): the axis is a tissue-independent squamous-differentiation marker.

Clinical significance: identity, not outcome (an honest negative). A reproducible axis need not be prognostic. In TCGA-BRCA (n = 866, 132 events) the basal score is not associated with overall survival (Cox HR 1.01, $p = 0.89$; adjusted for proliferation $p = 0.36$; KM log-rank $p = 0.29$), whereas the proliferation score is ($p = 0.001$). The basal axis does mark ER-negative/basal-like disease (AUROC 0.70). The discovered axis therefore captures lineage *identity* rather than outcome — reported plainly, because the method's value is finding real, robust biology, not inflated clinical claims.

3. Discussion

The method instantiates and unifies three established traditions — clinical-offset / incremental-value modelling (an established model as offset, omics on the residual; the closest methodological twin) [5]; biologically-informed models that bake known biology into structure [6,7]; and machine learning on established gene signatures [8]. Its specific contribution is a packaged, never-below-anchor, margin-gated residual on a *zero-parameter* prior, plus a residual-discovery framing with an explicit noise control, and a demonstration that the discovered axes' external reproducibility is predictable from their biological coherence. The honest scope: predictive gains over a strong anchor are modest — the value is *routing and discovery*, not large AUROC wins — consistent with the rarity of genuine super-additive multi-omics gains [1,2]. Discovered axes are candidate hypotheses; the basal axis is externally validated, the HER2 axis is cohort-specific, and other endpoints await further cohorts.

4. Methods

Anchored integration. `select_anchor` scores each modality by repeated stratified-CV AUROC (tie-broken by robustness) and picks the top composite, inside the outer CV. `anchored_integrate` pins the anchor and adds a secondary on its residual as `logit(anchor) + β·secondary`, with $\beta \geq 0$ chosen on held-out data ($\beta = 0$ allowed; a margin gate plus repeated inner CV control small-sample noise). `forward_integrate` / `auto_integrate` extend this to any number of modalities by greedy forward selection. **Knowledge anchor.** `signature_score` builds a fixed mean-z score from a curated gene set; `knowledge_anchored_integrate` pins it as the anchor. **Residual**

discovery. `anchored_residual_discovery` ranks features by partial correlation with the label controlling for the anchor, retains those orthogonal to the anchor ($|r| < 0.6$), gates the panel, and tests it against matched random panels and (for verification) held-out splits and permuted labels. Data: TCGA-BRCA (Xena HiSeqV2, HumanMethylation450, clinical) and METABRIC (microarray, clinical). Pathway enrichment: g:Profiler. All functions are in `omniomics.multiomics`; runners and recorded metrics are in the repository. **Scalability.** The discovery is linear and streams: a vectorized residualize-then-correlate, an optional one-pass Sure-Independence-Screening prefilter, Benjamini–Hochberg FDR, and parallel nulls let it run on genome-wide feature matrices out of core. As an end-to-end check, screening all 485,577 HumanMethylation450 probes ($n = 829$) to the top 5,000 by association with ER status took ≈ 24 s in a single low-memory pass; the ESR1-methylation anchor (AUROC 0.65) then rose to 0.90 with genome-wide CpGs, and the top CpG beyond ESR1 mapped to PGR (the canonical ER-coregulated gene) — textbook biology recovered at scale.

5. Data and Code Availability

Code, runners, recorded metrics, unit tests and continuous-integration guards: the `omniomics` repository (`reports/dmoi_*.py`, `*_results.csv`, `tests/`). Key runners: `dmoi_knowledge_anchor.py`, `dmoi_residual_discovery.py`, `dmoi_discovery_{er,her2}.py`, `dmoi_external_{metabric,her2_metabric}.py`, `dmoi_external_lung.py`, `dmoi_external_hnsc.py`, `dmoi_discovery_nsclc_io.py`, `dmoi_clinical_survival.py`, `dmoi_meth_genomewide.py` (genome-wide scale). Scale utilities: `omniomics.scale`, `omniomics.representations` Full methods note: `reports/anchored_integration_methods.md`. All data are public and de-identified: TCGA (BRCA, LUAD, LUSC, HNSC) via UCSC Xena; METABRIC via cBioPortal; the NSCLC anti-PD-1 cohort from Hellmann et al. (MSK 2018) via cBioPortal.

Declarations

Competing interests. The author declares no competing interests.

Funding. No external funding was received for this work.

Author contributions. H.R.K. conceived the method, performed all analyses, and wrote the manuscript.

Ethics. This study uses only publicly available, de-identified data from established repositories (TCGA/UCSC Xena, cBioPortal); no new human-subjects data were collected, so no additional ethics approval was required.

AI assistance. Analyses and drafting were carried out with the assistance of an AI coding agent under the author's direction; all results are reproducible from the cited runners.

Figure

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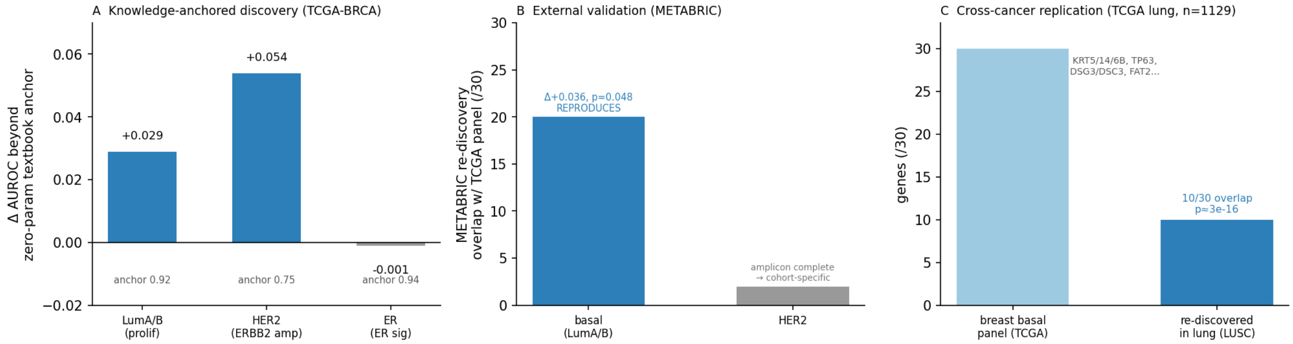

Figure 1: **Figure 1. Knowledge-anchored discovery and its external validation. (A)** Across three TCGA-BRCA endpoints, the gain in AUROC from adding genome-wide data beyond a zero-parameter textbook anchor: the data adds a real axis where the textbook is incomplete (LumA/B proliferation → basal, +0.029; HER2 amplicon → neuroendocrine/immune, +0.054) and nothing where it is complete (ER signature, −0.001; specificity). **(B)** In the independent METABRIC cohort, an unbiased re-discovery recovers the basal axis (20/30 overlap with the TCGA panel; direct replication Δ +0.036, $p = 0.048$) but not the HER2 axis (2/30), because the amplicon anchor is already complete in METABRIC. **(C)** Cross-cancer replication: in TCGA lung (LUAD vs LUSC, $n = 1{,}129$) the same proliferation-anchored residual re-discovers the squamous/keratinization program, overlapping the breast basal panel 10/30 (KRT5/14/6B, TP63, DSG3/DSC3, FAT2…; hypergeometric $p \approx 3\times10^{-16}$) — the discovered biology, not a cohort artefact.

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